

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Cent AI

Jun 2026 – present

Software Engineering Intern: Python, Medical AI, ONNX, TinyGrad, Kafka, Redis, AWS, RDP

Bengaluru

- Built and maintained an **ETL pipeline** connecting to **Neorad** to fetch, transport, and prepare **MRI scans** for downstream segmentation and reporting workflows, owning the path end-to-end from day one.
- Integrated an **organ segmentation model** with built-in **organ-sanity checks**, flagging **physical irregularities** and malformed output, and automating organ segmentation to reduce examination turnaround time by **25%**.
- Trained a **brain-age prediction model from scratch** on structural MRI to estimate the **brain-age gap** between predicted and chronological age across age-stratified cohorts.
- Built a dedicated **AI output evaluation system** for segmentation and report-verification results, making **model quality measurable, reproducible, and reviewable** in production.

Bentham AI

Dec 2025 – Feb 2026

Backend Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Built a resilient **Puppeteer and Node.js automation engine** for automating **legal and compliance workflows** around **MCA company-registration filings**, with API lookups, retries, session recovery, validation checks, and operator checkpoints; reduced manual filing time by **85%**.
- Implemented **structured draft generation** for company-name suggestions, descriptions, and filing fields, with **inspectable outputs** and **deterministic validation** before execution.
- Containerized services with **Docker**, deployed them on **Google Cloud Run**, and automated releases with **GitHub Actions**, improving deployment velocity by **70%**.

Google Summer of Code

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, NumPy, PyTorch, ONNX, TinyGrad

Remote

- Built **medical-imaging segmentation workflows** for **95 anatomical brain regions**, integrating **MRI preprocessing, PyTorch and TorchScript, ONNX and TinyGrad** inference, generated masks, and 3D inspection paths.
- Worked with **NIfTI and MGZ volumes**, voxel conformation, **orientation-sensitive preprocessing**, label remapping, LUT mapping, binary masks, and **QC checks** for model output correctness.
- Reduced MRI-processing runtime by **87%** through **asynchronous and parallel execution** while validating orientation, labels, **GPU and CPU** paths, and reproducible outputs.

Open Source

[Contributions](#) 

- Contributed maintainer-reviewed patches across **Invesalious, Kuadrant MCP Gateway, Tiled, CRIU, and VideoLAN**, spanning medical-data correctness, backend failover, parser robustness, security validation, and compatibility fixes.

Projects

Postificus Content Distribution Engine | *Go, Echo, PostgreSQL, Redis, RabbitMQ, Go-Rod*

[GitHub](#) 

- Solved fragmented publishing across Medium, Dev.to, and LinkedIn with a centralized **Go/Echo engine** built around **Go-Rod browser automation** and AI-driven content transformation, turning one draft into platform-ready posts.
- Engineered distributed execution for long-running browser jobs with **RabbitMQ retries and dead-letter queues**, **Redis** autosave, **PostgreSQL** persistence, circuit breakers, health checks, and **Prometheus** metrics to isolate third-party failures.

Hyoka Agent Reliability Layer | *Python, FastAPI, SQLAlchemy, PostgreSQL, OpenTelemetry*

[GitHub](#) 

- Solved unreliable AI-agent behavior without visibility, evaluation, or rollback by building Hyoka, a **reliability control plane** ingesting **OTLP/OpenInference** traces and replaying agent workflows for evaluation.
- Designed **worker-owned replay and evaluation execution with database leases**, an evaluator registry, and failure mining that converts failures into evidence-hashed candidate changes, validated against prior behavior before gated promotion, with an **SQLAlchemy/Alembic**-backed metadata model across SQLite and PostgreSQL.

Technical Skills

Languages: Python, Go, JavaScript, TypeScript

Frameworks: FastAPI, Echo, SQLAlchemy, Node.js, PyTorch

Tools: PostgreSQL, Redis, RabbitMQ, Docker, GCP, GitHub Actions, OpenTelemetry, Prometheus